

RAJARATA UNIVERSITY OF SRI LANKA

Department of Information and Communication Technology

ICT1209 – Web Technologies

Mini Project – Final Report

QR CODE BASED STUDENT ATTENDANCE MANAGEMENT SYSTEM

Theme: Smart Campus Automation

Group Members

R.M.P. Nisansala — ITT/2024/075 (2760)

L.P.M. Kavindi — ITT/2024/055 (2740)

Date: 26.09.2026

Contents

List Of Tables	3
List Of Figures.....	3
1.PROJECT OVERVIEW	4
1.1 Project Title	4
1.2 Theme	4
1.3 Purpose	4
1.4 Target Audience	4
1.5 Key Features	4
1.6 Technology Stack	4
2. SYSTEM DESIGN	5
2.1 System Architecture	5
2.2 Database Schema	6
2.3 Folder Structure	7
3. IMPLEMENTATION	7
3.1 Home Page (index.php)	7
3.2 Login Page (login.php).....	8
3.3 Register Page (register.php).....	8
3.4 Dashboard (dashboard.php).....	9
3.5 Contact Page (contact.php)	10
3.6 Database (phpMyAdmin)	10
3.7 Key Code Snippets	13
4. CHALLENGES AND SOLUTIONS	14
5. INDIVIDUAL CONTRIBUTION	15
5.1 Person A (ITT/2024/075).....	15
5.2 Person B (ITT/2024/055)	16
6. SETUP INSTRUCTIONS	17
6.1 Requirements.....	17
6.2 Installation Steps.....	17
Step 1: Install XAMPP	17
Step 2: Copy Project.....	17
Step 3: Import Database	17
Step 4: Configure Database	17
Step 5: Run the Application	17
7. FUTURE ENHANCEMENTS	18
8. REFERENCES	18
DECLARATION.....	18

List Of Tables

Table 1: User	4
Table 2: Features	4
Table 3: Used Technology	4
Table 4: Challenges and Solutions	16
Table 5: Work Contribution (2760)	16
Table 6: Work Contribution (2740)	17
Table 7: Example	18
Table 8: members	19

List Of Figures

Figure 1: System Architecture	5
Figure 2 : Database Schema	6
Figure 3 :Folder Structure	7
Figure 4 : Home Page	8
Figure 5 : Login Page	8
Figure 6 : Register Page	9
Figure 7: Lecturer Dashboard	10
Figure 8: Student Dashboard	10
Figure 9: Contact Page	11
Figure 10: User (database)	11
Figure 11: Attendance (database)	12
Figure 12: Sessions (database)	12
Figure 13: Messages (database)	13

1.PROJECT OVERVIEW

1.1 Project Title

QR Code Based Student Attendance Management System

1.2 Theme

Smart Campus Automation – Using QR technology to streamline attendance tracking in educational institutions.

1.3 Purpose

To develop a secure, efficient, and user-friendly web application that automates the student attendance process using QR code technology. Lecturers generate unique QR codes for each class session, and students scan these codes to mark their attendance instantly.

1.4 Target Audience

User Type	Description	Needs
Lecturers	Academic staff conducting classes	Generate QR codes, view attendance reports
Students	Learners attending classes	Scan QR codes, view attendance history

Table 1: User

1.5 Key Features

Feature	Description
User Authentication	Secure registration and login with password hashing
QR Code Generation	Lecturers generate unique QR codes for each session
QR Code Scanning	Students scan QR codes to mark attendance
Attendance History	View personal attendance records
Contact Form	Users can send messages
Dashboard	Role-based dashboard for students and lecturers

Table 2: Features

1.6 Technology Stack

Technology	Purpose
HTML5	Content structure
CSS3	Styling and layout
Bootstrap 5	Responsive UI framework
JavaScript	Client-side interactivity
PHP 8	Server-side scripting
MySQL	Database management

phpMyAdmin	Database GUI
XAMPP	Local development environment
Git & GitHub	Version control

Table 3: Used Technology

2. SYSTEM DESIGN

2.1 System Architecture

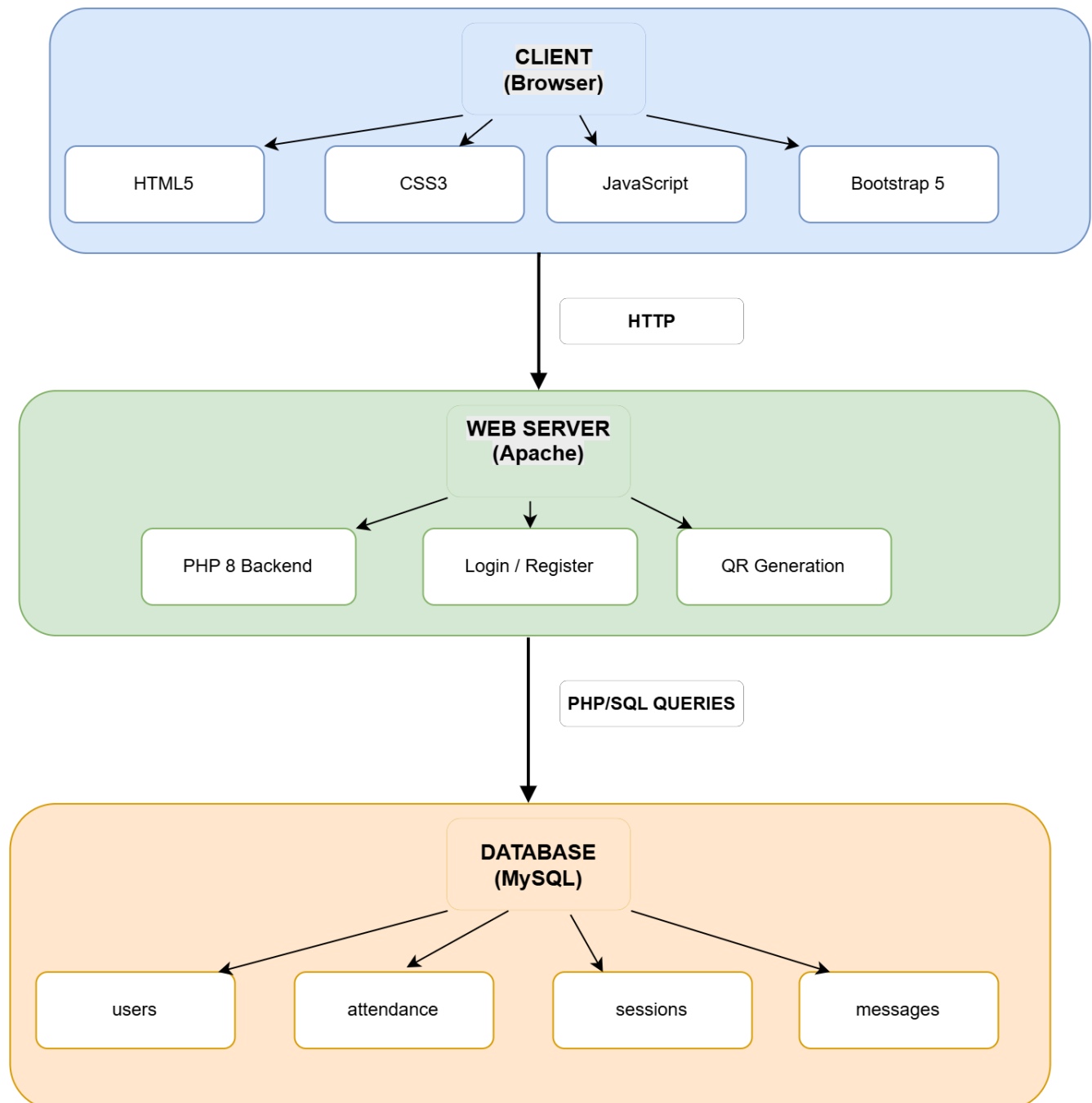


Figure 1: System Architecture

2.2 Database Schema

users (1) — (M) attendance

users (1) — (M) sessions

messages (independent table)

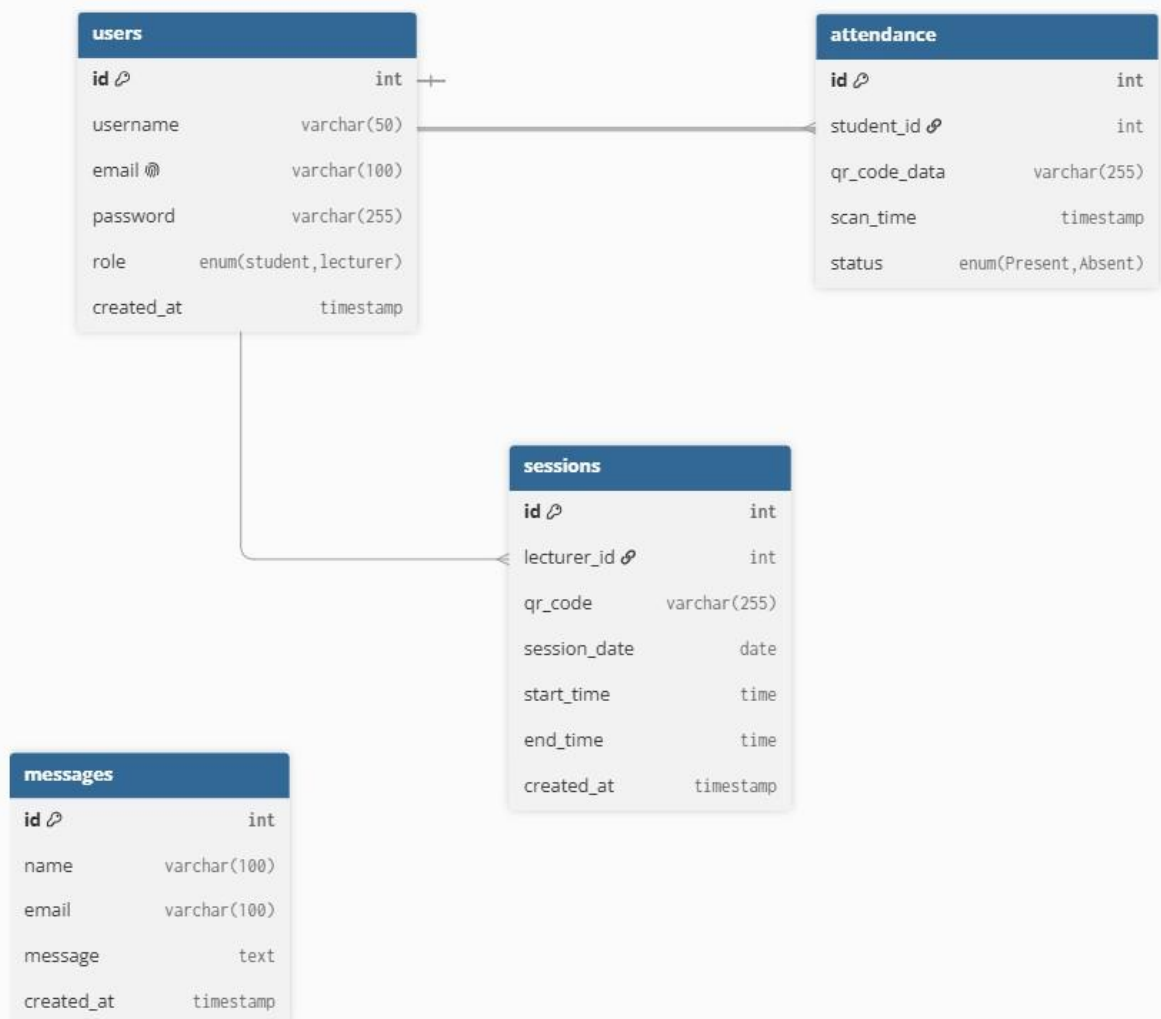


Figure 2: Database Schema

2.3 Folder Structure

```
qr_attendance/
├── auth/
│   ├── login.php
│   ├── register.php
│   └── logout.php
├── css/
│   └── style.css
├── js/
│   └── app.js
├── images/
├── Includes/
│   ├── db.php
│   └── functions.php
├── sql/
│   └── database.sql
├── index.php
├── dashboard.php
├── contact.php
├── index.html
├── login.html
├── register.html
├── dashboard.html
├── contact.html ─┘
├── README.md
└── database.sql
```

Figure 3 : Folder Structure

3. IMPLEMENTATION

3.1 Home Page (index.php)

Description: The landing page of the system with a navigation bar, hero section, three feature cards, and footer. It dynamically shows Login/Register buttons or Dashboard/Logout based on session status.

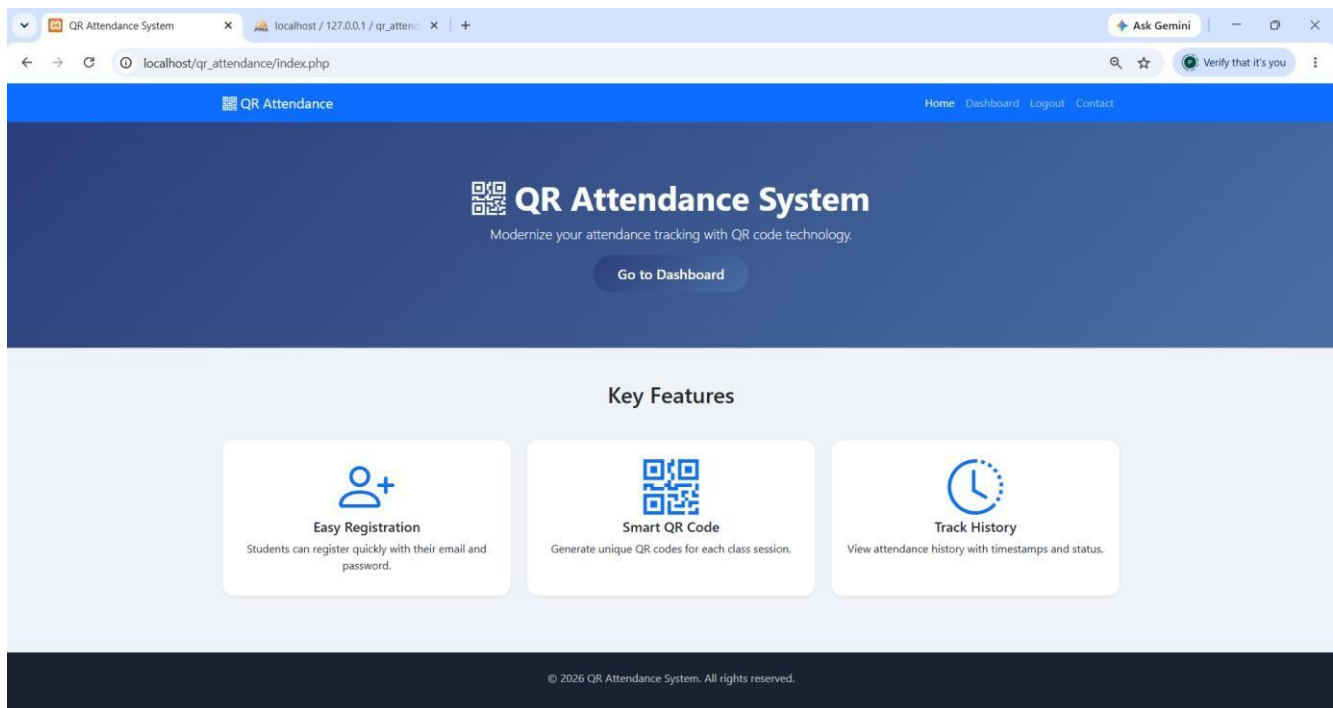


Figure 3: Home Page

3.2 Login Page (login.php)

Description: Allows registered users to log in using email and password. Uses password_verify() to validate hashed passwords and starts a session on success.

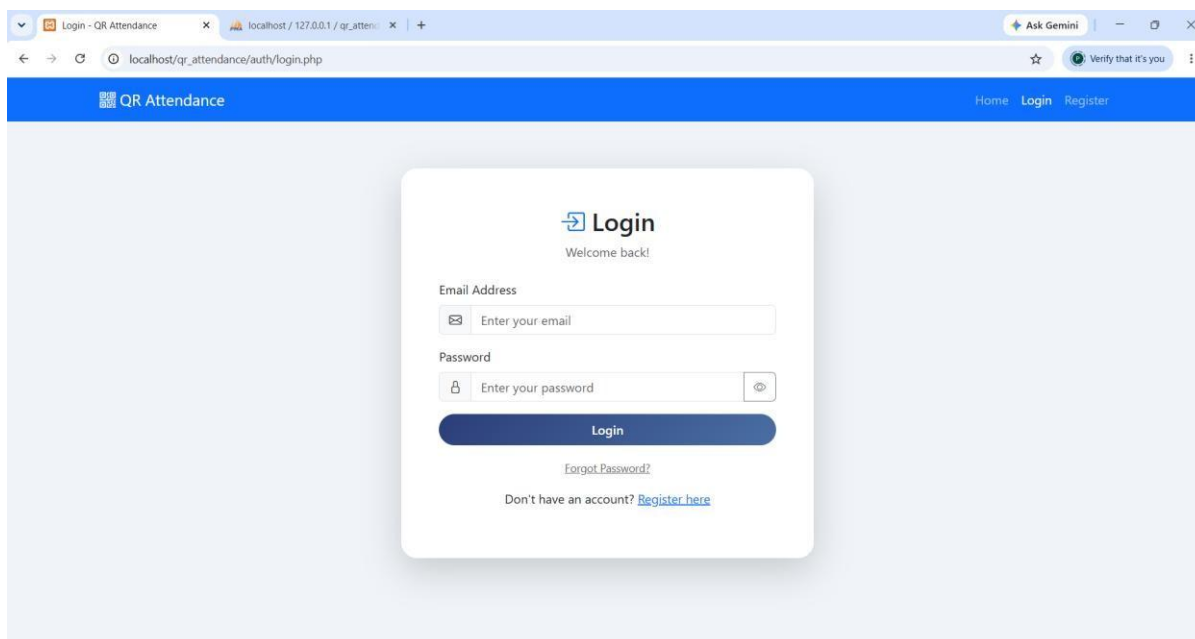


Figure 4: Login Page

3.3 Register Page (register.php)

Description: Allows new users to create an account. Validates email format, password length (minimum 6 characters), and confirms password match. Hashes the password with password_hash() before storing it in the database.

The screenshot shows the 'Register' page of the QR Attendance system. The page has a blue header with the logo and navigation links: Home, Login, Register, Dashboard, and Contact. The main content area is white and contains a registration form. The form has the following fields:

- Full Name:** A text input field with a placeholder 'Enter your full name'.
- Email Address:** A text input field with a placeholder 'Enter your email'.
- Password:** A text input field with a placeholder 'Create password (min 6 chars)'. Below it, a note says 'Password must be at least 6 characters'.
- Confirm Password:** A text input field with a placeholder 'Confirm your password'.

Below the form is a blue button labeled 'Register'. At the bottom of the form, there is a link: 'Already have an account? [Login here](#)'. The footer of the page is dark blue and contains the text: '© 2026 QR Attendance System. All rights reserved.'

Figure 5: Register Page

3.4 Dashboard (dashboard.php)

Description: Role-based dashboard. Student users can scan QR codes to mark attendance and view attendance history. Lecturer users can generate QR codes for sessions and view class attendance.

1. Lecturer Dashboard

The screenshot shows the 'Lecturer Dashboard' of the QR Attendance system. The page has a blue header with the logo and navigation links: Home, Ajith Bandara (Lecturer), and Logout. The main content area is white and contains the following elements:

- User Profile:** A card showing the user's name 'Ajith Bandara' and role 'Lecturer'. It also displays '100%' Attendance and '3' Sessions.
- Quick Links:** A section with two links: 'Generate QR' and 'History'.
- Generate QR Code:** A button labeled 'Generate New QR'.
- Class Attendance Summary:** A table showing the attendance of students in a class. The table has the following data:

Student	Date	Time	Status
Piyumi Nisansala	27 Sep 2026	03:18 PM	Present

Below the table is a red button labeled 'Download PDF'. The footer of the page is dark blue and contains the text: '© 2026 QR Attendance System'.

Figure 6: Lecturer Dashboard

2.Student Dashboard

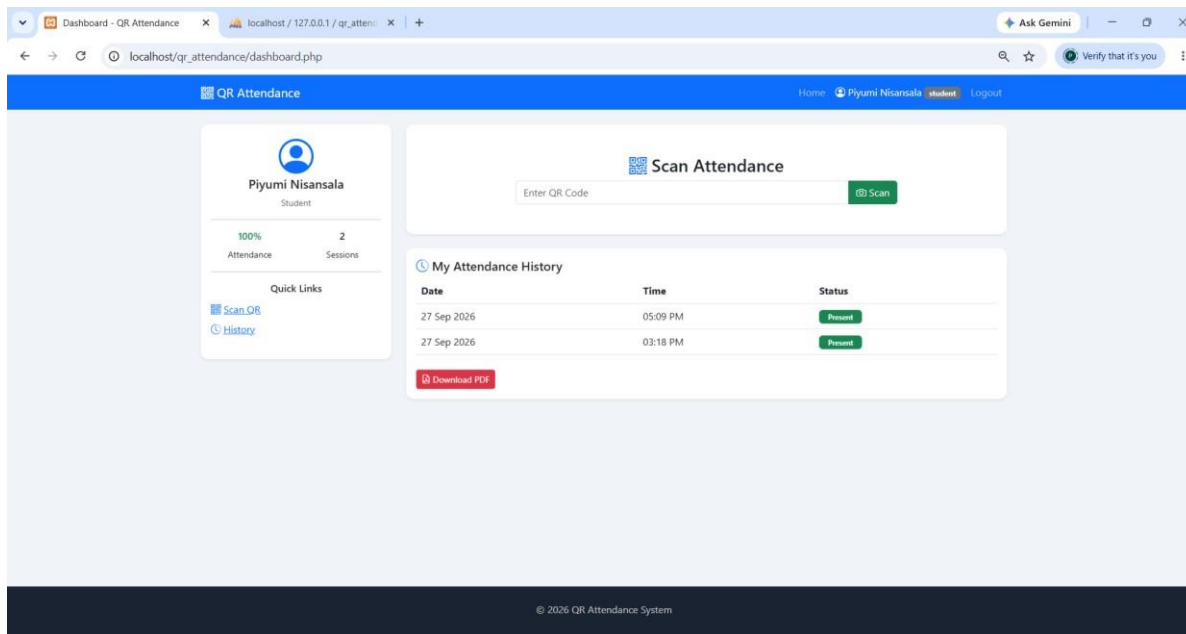


Figure 7: Student Dashboard

3.5 Contact Page (contact.php)

Description: Allows visitors to send messages. Form data is validated and stored in the messages table using prepared statements.

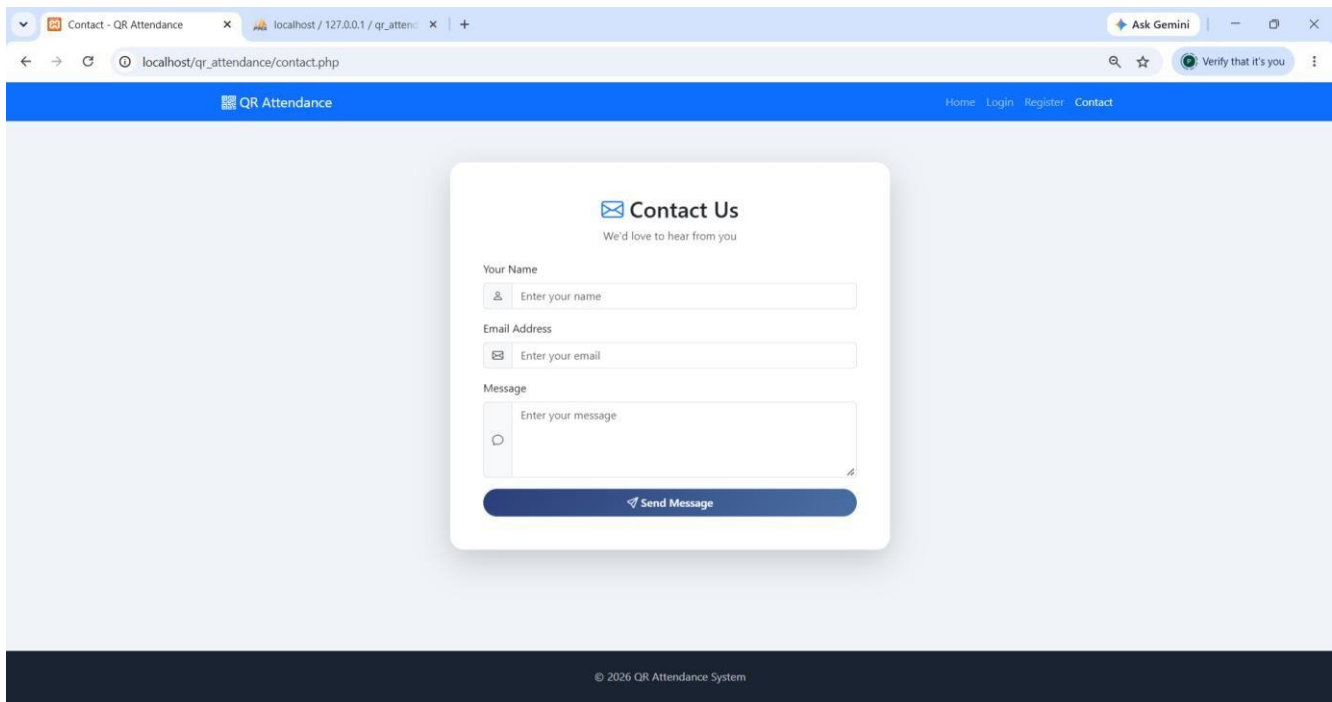


Figure 8: Contact Page

3.6 Database (phpMyAdmin)

Description: MySQL database qr_attendance with four tables: users, attendance, sessions, and messages. Passwords are securely stored as BCrypt hashes (starting with \$2y\$10\$). 1.User Table

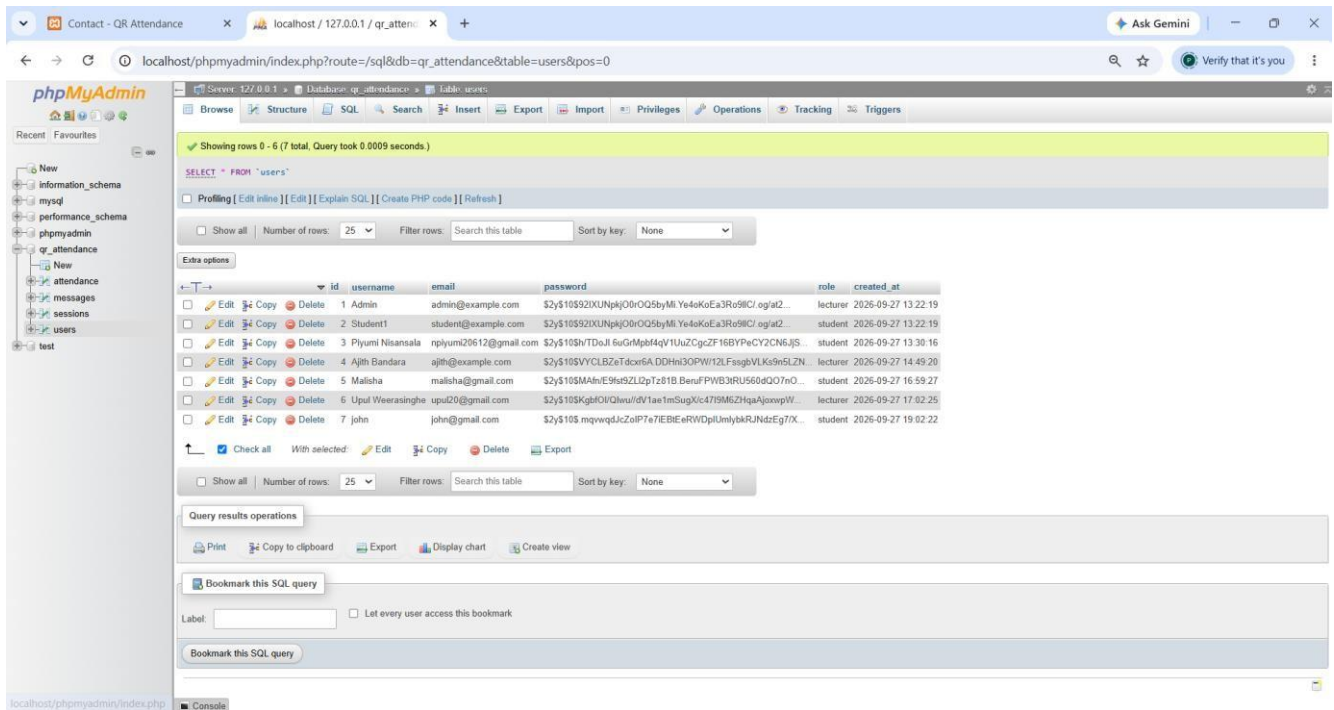


Figure 9: User (database)

2.Attendance Table

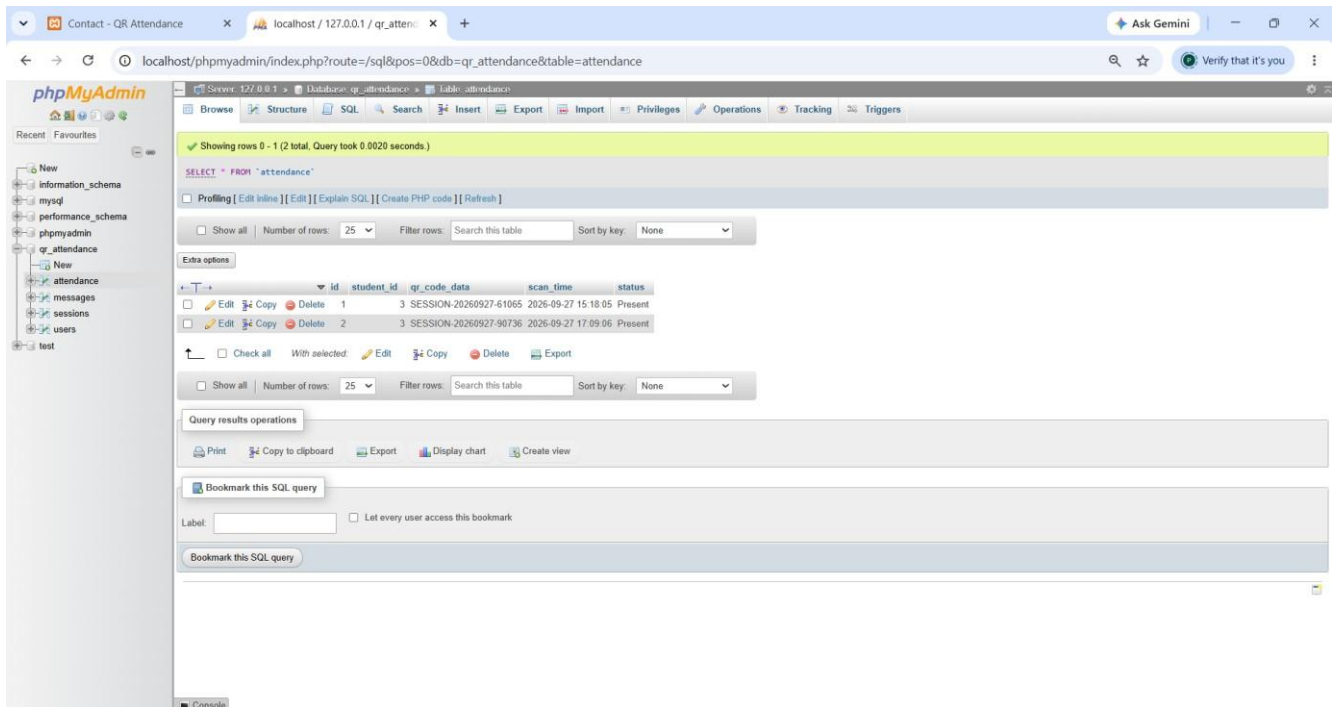


Figure 10: Attendance (database)

3.Sessions Table

Showing rows 0 - 3 (4 total, Query took 0.0013 seconds)

SELECT * FROM 'sessions'

Number of rows: 25

	id	lecturer_id	qr_code	session_date	start_time	end_time	created_at
☐	1	4	SESSION-20260927-97462	2026-09-27	11:43:45	12:43:45	2026-09-27 15:13:45
☐	2	4	SESSION-20260927-61065	2026-09-27	11:47:02	12:47:02	2026-09-27 15:17:02
☐	3	4	SESSION-20260927-33293	2026-09-27	11:59:16	12:59:16	2026-09-27 15:29:16
☐	4	6	SESSION-20260927-90736	2026-09-27	13:38:40	14:38:40	2026-09-27 17:08:40

Figure 11: Sessions (database)

4.messages Table

Showing rows 0 - 0 (1 total, Query took 0.0026 seconds)

SELECT * FROM 'messages'

Number of rows: 25

	id	name	email	message	created_at
☐	1	Piyumi Nisansala	npiyumi20612@gmail.com	Send Message	2026-09-27 13:58:54

Figure 12: Messages (database)

3.7 Key Code Snippets

Database Connection (Includes/db.php)

```
<?php
$host = 'localhost';
$dbname = 'qr_attendance';
$username = 'root';
$password = '';
try
{
    $conn = new PDO ("mysql: host=$host; dbname=$dbname", $username, $password);
    $conn->setAttribute (PDO: ATTR_ERRMODE, PDO: ERRMODE_EXCEPTION);
    $conn->setAttribute (PDO: ATTR_DEFAULT_FETCH_MODE, PDO: FETCH_ASSOC);
} catch (PDOException $e) {      die ("Connection
failed: " . $e->getMessage ());
} if (session_status () ===
PHP_SESSION_NONE) {      session_start ();
}
?>
```

User Registration

```
$hashed_password = password_hash($password, PASSWORD_DEFAULT); $stmt = $conn->prepare
("INSERT INTO users (username, email, password) VALUES (?, ?, ?)");
$stmt->execute ([$username, $email, $hashed_password]);
```

User Login

```
$stmt = $conn->prepare ("SELECT * FROM users WHERE email = ?");
$stmt->execute([$email]);
$user = $stmt->fetch ();
if ($user && password_verify ($password, $user['password']))
{
    session_regenerate_id(true);
    $_SESSION['user_id'] = $user['id'];
    $_SESSION['username'] = $user['username'];
    $_SESSION['role'] = $user['role'];
    header ('Location: ../dashboard.php');
}
```

QR Code Generation (Lecturer)

```
$qr_code = 'SESSION-' . date('Ymd') . '-' . rand (10000, 99999);
$stmt = $conn->prepare ("INSERT INTO sessions (lecturer_id, qr_code, session_date, start_time,
end_time) VALUES (?, ?, ?, ?, ?) ");
$stmt->execute ([$user_id, $qr_code, $session_date, $start_time, $end_time]);
```

QR Code Scanning (Student)

```
$stmt = $conn->prepare ("SELECT id FROM sessions WHERE qr_code = ? AND session_date = CURDATE
()");
$stmt->execute([$scanned_code]);
if ($stmt->rowCount () > 0)
{
    $stmt = $conn->prepare ("INSERT INTO attendance (student_id, qr_code_data, status) VALUES (?
?, 'Present')");
    $stmt->execute ([$user_id, $scanned_code]);
}
```

Contact Form

```
$stmt = $conn->prepare ("INSERT INTO messages (name, email, message) VALUES (?, ?, ?)"); $stmt->execute ([$name, $email, $message]);
```

4. CHALLENGES AND SOLUTIONS

Phase	Challenge	Solution
Phase 1	Choosing between the provided theme and custom QR idea	Prepared detailed proposal justifying theme, target audience, and features
Phase 1	No experience with wireframe design tools	Used Figma, free templates and hand-drawn sketches that were digitized
Phase 1	Setting up GitHub repository structure.	Followed PHP project structure guidelines and used official. gitignore
Phase 2	Bootstrap CDN not loading, navbar invisible	Moved Bootstrap <link> to be first item in <head> before custom CSS
Phase 2	Typo errors in HTML (navabar-collapse, col=lg8, type="botton")	Carefully reviewed HTML, corrected typos, tested with Live Server
Phase 2	Form validation missing on client side	Added JavaScript validation in js/app.js for empty fields and email format
Phase 2	Password fields had no show/hide option	Added eye icon using Bootstrap Icons and togglePassword () JS function
Phase 2	Project outside XAMPP htdocs caused 404 error	Copied project into C:\xampp\htdocs\qr_attendance\
Phase 2	Live Server (HTML) vs XAMPP (PHP) conflict	Used Live Server for HTML/CSS and XAMPP for PHP testing
Phase 3	MySQL failed to start repeatedly ("shutdown unexpectedly")	Stopped conflicting Windows services, deleted corrupt ib_logfile0 and ib_logfile1, ran XAMPP as Admin
Phase 3	phpMyAdmin refused to connect (ERR_CONNECTION_REFUSED)	Cleared browser cache, flushed DNS with ipconfig /flushdns, accessed via localhost/phpMyAdmin
Phase 3	Port 3306 conflict with another MySQL instance	Stopped conflicting service via Windows Task Manager → Services

Phase 3	Unknown database 'qr_attendance'" error on login	Manually created database in phpMyAdmin and imported database.sql
Phase 3	Undefined function getUserId ()" crashed dashboard	Added getUserId (), getUsername (), getUserRole () to includes/db.php
Phase 3	"Undefined variable \$attendance_records" error	Re-inserted attendance query block in dashboard.php
Phase 3	"Call to undefined function isLoggedIn ()"	Added isLoggedIn () helper to includes/db.php
Phase 3	Session data not persisting after login	Added session_start () in db.php and session_regenerate_id(true) after login
Phase 3	PASSWORD_DEFAULT did not match the guide's PASSWORD_BCRYPT requirement	Changed password_hash (\$password, PASSWORD_DEFAULT) to PASSWORD_BCRYPT
Phase 3	Lecturer could see other lecturers' attendance data	Added JOIN sessions s ON a. qr_code_data = s. qr_code WHERE s. lecturer_id =? to filter by lecturer
Phase 3	Duplicate attendance records for same session	Added check SELECT id FROM attendance WHERE student_id =? AND qr_code_data =? before insert
Phase 3	Lecturer could see other lecturers' attendance data	Added JOIN sessions s ON a. qr_code_data = s. qr_code WHERE s. lecturer_id =? to filter by lecturer
Phase 3	GitHub push rejected ("Updates were rejected")	Ran git pull origin main --rebase then pushed again
Phase 3	Apache port 80 conflict with another program	Changed Apache to port 8080 in httpd.conf
Phase 3	database.sql was excluded by .gitignore	Removed *.sql pattern and used git add -f sql/database.sql
Phase 3	PowerShell terminal stuck with >> waiting prompt	Pressed Ctrl + C to cancel and re-ran commands separately
Phase 3	Git showed "deleted" files after restructuring folders	Ran git add. to stage moves and used refactor: commit message

Table 4: Challenges and Solutions

5. INDIVIDUAL CONTRIBUTION

5.1 Person A (ITT/2024/075)

Phase	Task	Status
Phase 1	Project idea, theme, purpose, goals	✓

Phase 1	Target audience, features list	✓
Phase 1	Technology stack, proposal writing	✓
Phase 1	GitHub setup, wireframes (Home, Login, Register)	✓
Phase 2	index.html, login.html, register.html	✓
Phase 2	Base CSS, navbar, footer, hero, buttons, cards	✓
Phase 2	Form validation, event handling	✓
Phase 3	Database creation, users/attendance/sessions tables	✓
Phase 3	includes/db.php	✓
Phase 3	auth/register.php, auth/login.php, auth/logout.php	✓
Phase 3	index.php, database. SQL, Testing, Debugging	✓
Final Report	Cover page, project overview, system design	✓
Final Report	Database schema, ER diagram, folder structure	✓
Final Report	Frontend implementation, setup instructions	✓

Table 5: Work Contribution (2760)

5.2 Person B (ITT/2024/055)

Phase	Task	Status
Phase 1	Features list contribution	✓
Phase 1	Wireframes (Dashboard, Contact)	✓
Phase 2	dashboard.html, contact.html	✓
Phase 2	Dashboard styles, table/badge styles, responsive design	✓
Phase 2	Scan QR, Generate QR, Load Demo Data features	✓
Phase 3	messages table	✓
Phase 3	includes/functions.php	✓
Phase 3	contact.php, dashboard.php	✓
Final Report	Report writing and compilation	✓
Final Report	Backend implementation, screenshots	✓

Final Report	Challenges & solutions, references	✓
--------------	------------------------------------	---

Table 6: Work Contribution (2740)

6. SETUP INSTRUCTIONS

6.1 Requirements

- XAMPP (or WAMP)
- Web browser (Chrome/Firefox)
- Git (optional)

6.2 Installation Steps

Step 1: Install XAMPP

1. Download XAMPP from <https://www.apachefriends.org/>
2. Install with default settings.
3. Open XAMPP Control Panel.
4. Start Apache and MySQL.

Step 2: Copy Project

5. Copy the project folder to: C:\xampp\htdocs\qr_attendance\

Step 3: Import Database

6. Open browser: <http://localhost/phpmyadmin>
7. Click New → Database name: qr_attendance → Create.
8. Click Import tab → Choose database.sql → Click Go.

Step 4: Configure Database

9. Open: Includes/db.php
10. Update credentials if needed:

```
$host = 'localhost';
$dbname = 'qr_attendance';
$username = 'root';
$password = '';
```

Step 5: Run the Application

11. Open browser: http://localhost/qr_attendance/
12. Register a new account or login with:

Email	Password
admin@example.com	admin123
student@example.com	student123

Table 7: sample Login Credentials

7. FUTURE ENHANCEMENTS

The following features were planned but not implemented in this version:

1. Real Camera-Based QR Scanning: Currently, students enter the QR code manually. In production, this would integrate with the device camera using html5-qrcode.
2. Time-Limited QR Codes: Codes currently remain valid for the entire day. A future version would add a countdown timer (10-15 minutes).
3. Email Notifications: Verification emails and attendance notifications using PHPMailer or SMTP.
4. PDF Report Generation: Proper PDF export using FPDF or TCPDF library.
5. Administrator Dashboard: A dedicated admin panel for managing users and courses.
6. Parent Portal: Parents can monitor their child's attendance

8. REFERENCES

1. *Bootstrap 5 Documentation*. (2024). getbootstrap.com
2. *PHP Documentation*. (2024). [php.net](https://www.php.net)
3. *MySQL Documentation*. (2024). [mysql.com](https://www.mysql.com)
4. *W3Schools HTML/CSS Tutorial*. (2024). [w3schools.com](https://www.w3schools.com)
5. *MDN Web Docs*. (2024). developer.mozilla.org
6. *QR Code Technology*. (2024). [qrcode.com](https://www.qrcode.com)
7. *Bootstrap 5 Documentation*. (2024). <https://getbootstrap.com/>

DECLARATION

We hereby declare that this project is our own work and that both team members have contributed to its completion.

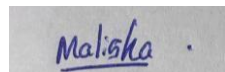
Name	Reg. No	Index No	Signature
R.M.P. Nisansala	ITT/2024/075	2760	
L.P.M. Kavindi	ITT/2024/055	2740	

Table 8: members